

LUNAR AUTONOMOUS LANDING ZONE ASSESSMENT & HABITAT PLANNING SYSTEM

PHYSICS-INFORMED SURVEY, HAZARD SCORING, ILLUMINATION ANALYSIS, AND HABITAT PLANNING FOR FUTURE LUNAR OPERATIONS



AUTONOMOUS SURVEY



TERRAIN ANALYSIS



ILLUMINATION ANALYSIS



COMMUNICATIONS ANALYSIS



HABITAT PLANNING



MISSION OPTIMIZATION



MISSION OVERVIEW

Mission ID	LZAP-2026-001
Target Body	MOON
Mission Profile	Landing Zone + Habitat Preparation
Date (UTC)	JUNE 2, 2026 14:32:10
System Status	NOMINAL

PLANETARY ENVIRONMENT

Body	Gravity (m/s ²)	Solar Flux (W/m ²)
Moon	1.62	1361
Mars	3.71	586
Phobos	0.0057	586
Europa	1.315	50
Custom	User-defined	User-defined

SYSTEM HEALTH

POWER	94%
COMMS	98%
SENSORS	95%
NAVIGATION	97%
AUTONOMY & GNC	96%

MAP LAYERS

☒ Terrain Elevation	☒
☒ Slope Map	☒
☒ Hazard Zones	☒
☒ Illumination (Sunlight)	☒
☒ Communication Range	☒
☒ Thermal Map	☒
☒ Proposed Habitat Zones	☒

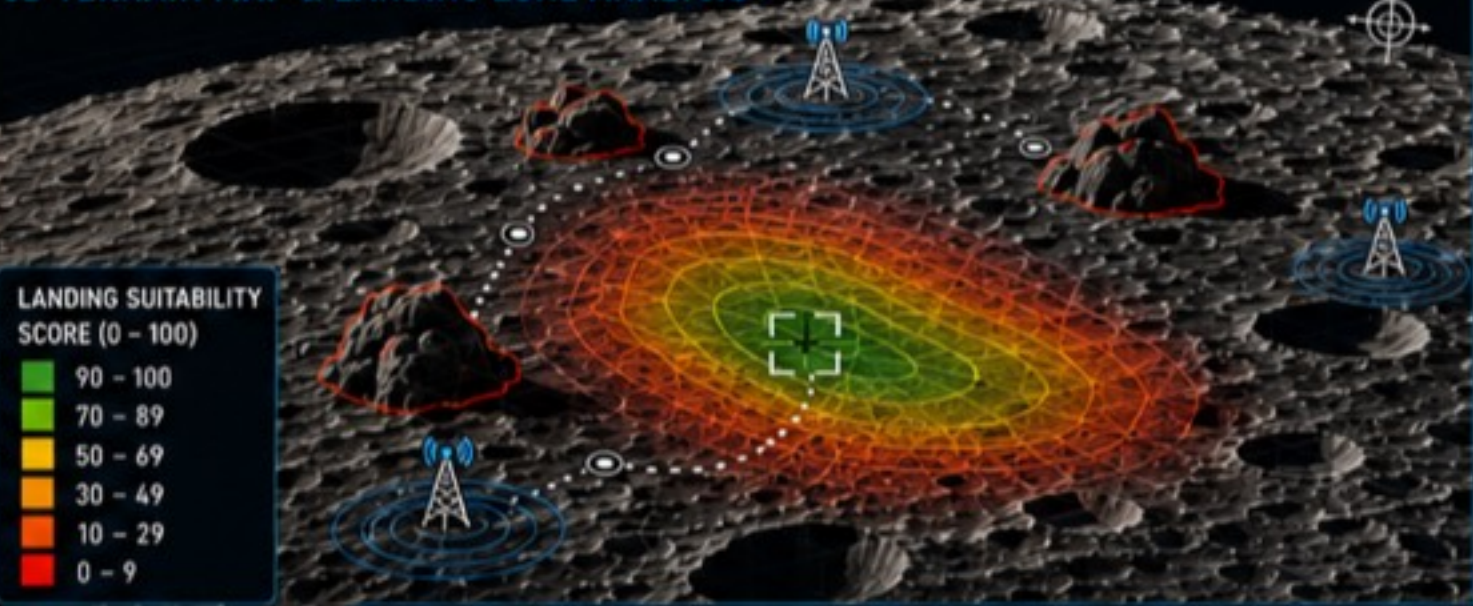
MISSION TIMELINE

1	2	3	4		5	6	8	9
MISSION CONFIG	TERRAIN GENERATION	ILLUMINATION ANALYSIS	COMMUNICATIONS ANALYSIS	HAZARD ASSESSMENT	LANDING ZONE SELECTION	ROVER SURVEY PLANNING	HABITAT RECOMMENDATION	MISSION EXPORT
T+ 00:00	T+ 0.3 HR	T+ 0.5 HR	T+ 0.7 HR	T+ 0.9 HR	T+ 1.1 HR	T+ 1.3 HR	T+ 1.6 HR	T+ 1.8 HR

OPTIONAL LLM ADVISORY LAYER

Status	ENABLED
Model	GPT-5.5-mini
Role	Advisory only

3D TERRAIN MAP & LANDING ZONE ANALYSIS



LANDING SUITABILITY SCORE (0 - 100)

90 - 100
70 - 89
50 - 69
30 - 49
10 - 29
0 - 9

ORBIT VIEW 2D MAP 3D MAP SUN VECTOR CROSS SECTION

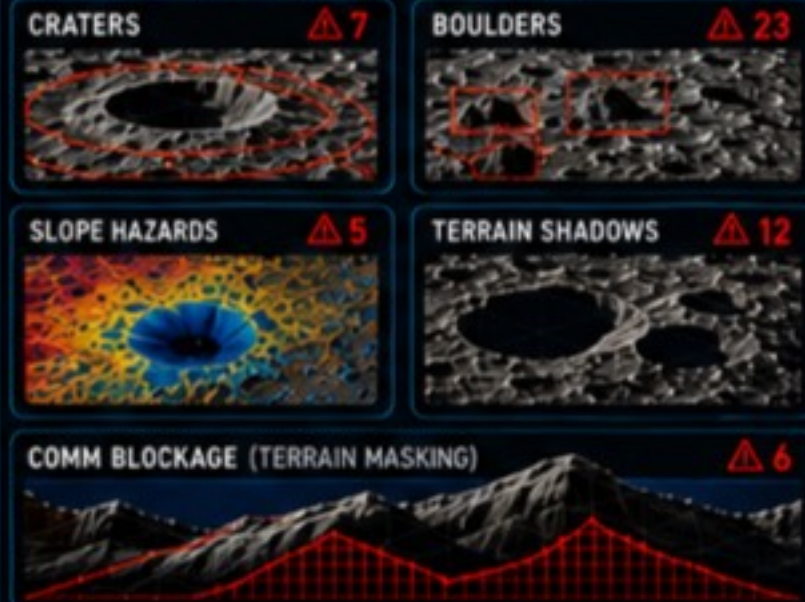
BEST LANDING ZONE

92 /100

- SLOPE: 2.1°
- TERRAIN SUITABILITY: HIGH
- BOULDER DENSITY: LOW
- HAZARDS: 3 SMALL
- ILLUMINATION: HIGH
- SOLAR FLUX: 1361 W/m²
- COMM INDEX: 0.86
- THERMAL ENV. INDEX: 0.90
- TERRAIN STABILITY: HIGH

RECOMMENDED

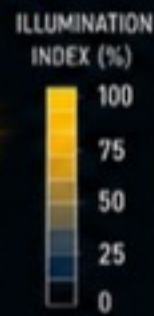
PHYSICS-BASED HAZARD ASSESSMENT



VIEW FULL HAZARD REPORT

ILLUMINATION ANALYSIS

SUN VECTOR GEOMETRY



SUN AZIMUTH	134.6°
SUN ELEVATION	31.2°
SHADOW FRACTION	18.6%
INCIDENCE ANGLE @ LZ	28.9°

COMMUNICATIONS ANALYSIS

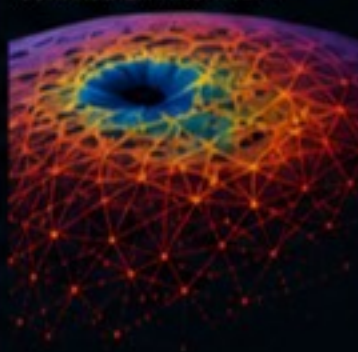
LINE-OF-SIGHT TO RELAY



COMM INDEX	0.86
LOS BLOCKED FRACTION	0.14
RELAY AZIMUTH	215.3°
RELAY ELEVATION	22.7°
DEEP-SPACE PENALTY	0.90

THERMAL ASSESSMENT

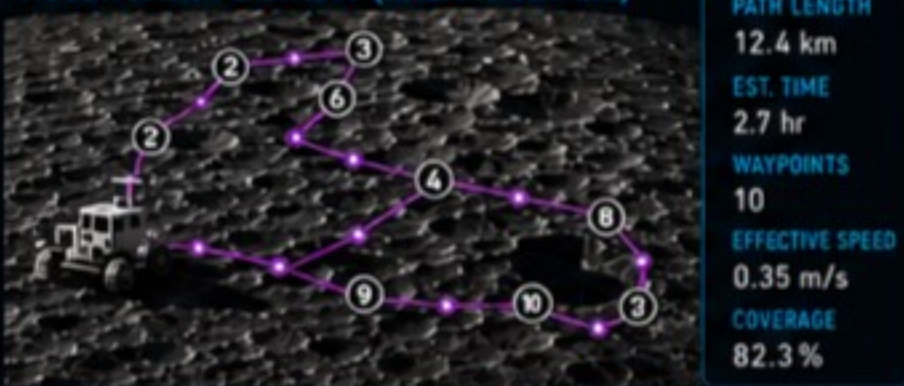
SURFACE ENVIRONMENT



THERMAL PENALTY	0.10
THERMAL ENV. INDEX	0.90
ENVIRONMENT SEVERITY	LOW

THERMAL MODEL ACTIVE

ROVER PATH PLANNING (HAZARD-AWARE)



PATH LENGTH	12.4 km
EST. TIME	2.7 hr
WAYPOINTS	10
EFFECTIVE SPEED	0.35 m/s
COVERAGE	82.3%

VIEW WAYPOINTS & PROFILE

HABITAT & INFRASTRUCTURE PLAN

- HABITAT ZONE
- POWER ZONE
- COMMS RELAY



CONTOURS: 10 m INTERVAL

RECOMMENDED LOCATIONS (m)

ELEMENT	X	Y	RADIUS
LANDING ZONE	-24.8	36.1	120 m
HABITAT ZONE	90.2	81.7	118 m
POWER ZONE	-134.2	121.3	161 m
COMMS RELAY	155.1	-39.0	179 m

MISSION SUMMARY

LANDING ZONE SCORE	RISK CLASSIFICATION	ILLUMINATION INDEX	COMM INDEX
92 /100	LOW RISK	86%	0.86
HAZARDS NEAR LZ	SURVEY TIME	SURVEY COVERAGE	THERMAL ENV. INDEX
3	2.7 hr	82.3%	0.90

MISSION PRODUCTS

- Mission Report (JSON)
- Mission Summary (CSV)
- Terrain & Hazard Maps (PNG)
- Recommended Coordinates (CSV)

EXPORT ALL PRODUCTS

SENSORS & INSTRUMENTS



Note: This application uses simplified models and surrogate metrics for concept development and decision support. Use mission-specific data, validated tools, and detailed analysis for flight-critical planning.

AUTONOMY | INTELLIGENCE | EXPLORATION | SYSTEM ENGINEERING

